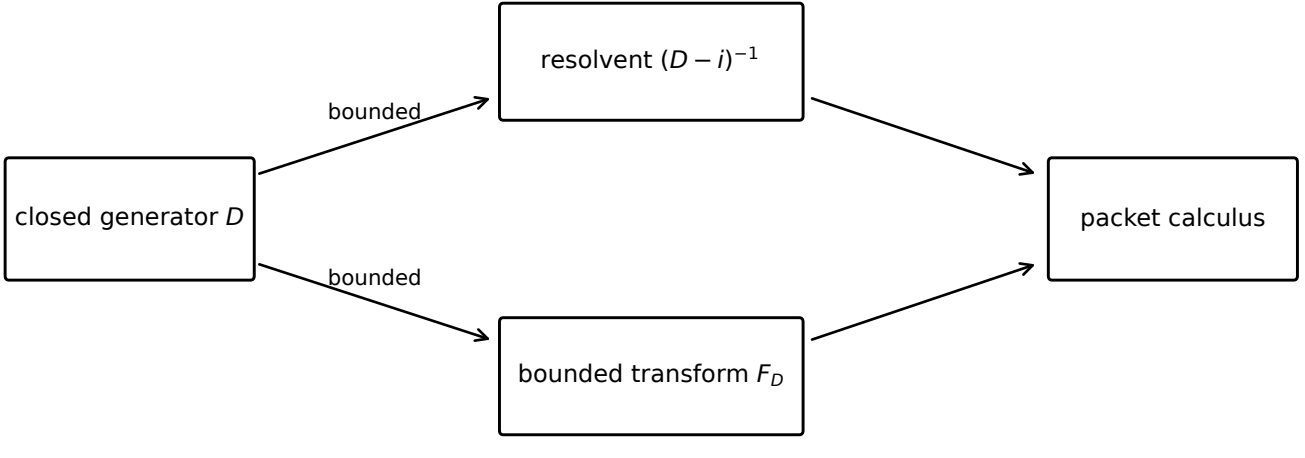unbounded structure enters through bounded surrogates and graph data